STE1:sigmoid t=0.1 + STE3:identity

10dB , 200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 96.51%
Recovered entity accuracy on unique test IDs: 87.11%
Recovered triple exact-match accuracy: 90.14%
Frequency-balanced triple recovery accuracy: 84.20%

seed = 41

Output dimension: 200
Recovered entity accuracy: 97.79%
Recovered entity accuracy on unique test IDs: 89.18%
Recovered triple exact-match accuracy: 93.56%
Frequency-balanced triple recovery accuracy: 93.49%

seed = 40

Output dimension: 200
Recovered entity accuracy: 97.59%
Recovered entity accuracy on unique test IDs: 88.66%
Recovered triple exact-match accuracy: 93.16%
Frequency-balanced triple recovery accuracy: 93.51%

seed = 39

Output dimension: 200
Recovered entity accuracy: 96.78%
Recovered entity accuracy on unique test IDs: 87.63%
Recovered triple exact-match accuracy: 90.34%
Frequency-balanced triple recovery accuracy: 89.80%

seed = 38

Output dimension: 200
Recovered entity accuracy: 87.59%
Recovered entity accuracy on unique test IDs: 90.72%
Recovered triple exact-match accuracy: 63.38%
Frequency-balanced triple recovery accuracy: 87.96%

seed = 37

Output dimension: 200
Recovered entity accuracy: 86.05%
Recovered entity accuracy on unique test IDs: 90.21%
Recovered triple exact-match accuracy: 59.56%
Frequency-balanced triple recovery accuracy: 83.29%

seed = 36

Output dimension: 200
Recovered entity accuracy: 97.85%
Recovered entity accuracy on unique test IDs: 89.69%
Recovered triple exact-match accuracy: 93.76%
Frequency-balanced triple recovery accuracy: 93.76%

seed = 35

Output dimension: 200
Recovered entity accuracy: 98.12%
Recovered entity accuracy on unique test IDs: 91.24%
Recovered triple exact-match accuracy: 94.77%
Frequency-balanced triple recovery accuracy: 94.69%

seed = 34

Output dimension: 200
Recovered entity accuracy: 85.65%
Recovered entity accuracy on unique test IDs: 85.57%
Recovered triple exact-match accuracy: 59.15%
Frequency-balanced triple recovery accuracy: 79.09%

seed = 33

Output dimension: 200
Recovered entity accuracy: 86.92%
Recovered entity accuracy on unique test IDs: 88.66%
Recovered triple exact-match accuracy: 61.57%
Frequency-balanced triple recovery accuracy: 83.37%

0dB, 200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 78.81%
Recovered entity accuracy on unique test IDs: 69.07%
Recovered triple exact-match accuracy: 41.65%
Frequency-balanced triple recovery accuracy: 52.87%

seed = 41

Output dimension: 200
Recovered entity accuracy: 77.53%
Recovered entity accuracy on unique test IDs: 64.43%
Recovered triple exact-match accuracy: 40.64%
Frequency-balanced triple recovery accuracy: 46.57%

seed = 40

Output dimension: 200
Recovered entity accuracy: 71.56%
Recovered entity accuracy on unique test IDs: 68.04%
Recovered triple exact-match accuracy: 34.41%
Frequency-balanced triple recovery accuracy: 48.31%

seed = 39

Output dimension: 200
Recovered entity accuracy: 74.51%
Recovered entity accuracy on unique test IDs: 71.13%
Recovered triple exact-match accuracy: 32.60%
Frequency-balanced triple recovery accuracy: 41.41%

seed = 38

Output dimension: 200
Recovered entity accuracy: 74.18%
Recovered entity accuracy on unique test IDs: 63.40%
Recovered triple exact-match accuracy: 29.78%
Frequency-balanced triple recovery accuracy: 39.49%

seed = 37

Output dimension: 200
Recovered entity accuracy: 73.91%
Recovered entity accuracy on unique test IDs: 65.98%
Recovered triple exact-match accuracy: 34.41%
Frequency-balanced triple recovery accuracy: 52.02%

seed = 36

Output dimension: 200
Recovered entity accuracy: 79.95%
Recovered entity accuracy on unique test IDs: 69.07%
Recovered triple exact-match accuracy: 44.87%
Frequency-balanced triple recovery accuracy: 54.99%

seed = 35

Output dimension: 200
Recovered entity accuracy: 76.66%
Recovered entity accuracy on unique test IDs: 69.07%
Recovered triple exact-match accuracy: 36.22%
Frequency-balanced triple recovery accuracy: 51.72%

seed = 34

Output dimension: 200
Recovered entity accuracy: 78.00%
Recovered entity accuracy on unique test IDs: 66.49%
Recovered triple exact-match accuracy: 39.44%
Frequency-balanced triple recovery accuracy: 42.17%

seed = 33

Output dimension: 200
Recovered entity accuracy: 73.78%
Recovered entity accuracy on unique test IDs: 67.53%
Recovered triple exact-match accuracy: 32.80%
Frequency-balanced triple recovery accuracy: 42.98%

STE1:identity + STE3:identity

10dB, 200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 80.01%
Recovered entity accuracy on unique test IDs: 80.41%
Recovered triple exact-match accuracy: 44.87%
Frequency-balanced triple recovery accuracy: 64.32%

seed = 41

Output dimension: 200
Recovered entity accuracy: 82.36%
Recovered entity accuracy on unique test IDs: 80.41%
Recovered triple exact-match accuracy: 49.90%
Frequency-balanced triple recovery accuracy: 62.91%

seed = 40

Output dimension: 200
Recovered entity accuracy: 84.64%
Recovered entity accuracy on unique test IDs: 82.47%
Recovered triple exact-match accuracy: 55.53%
Frequency-balanced triple recovery accuracy: 72.53%

seed = 39

Output dimension: 200
Recovered entity accuracy: 86.12%

Recovered entity accuracy on unique test IDs: 84.02%
Recovered triple exact-match accuracy: 60.36%
Frequency-balanced triple recovery accuracy: 81.72%

seed = 38

Output dimension: 200
Recovered entity accuracy: 94.23%
Recovered entity accuracy on unique test IDs: 80.41%
Recovered triple exact-match accuracy: 83.70%
Frequency-balanced triple recovery accuracy: 78.21%

seed = 37

Output dimension: 200
Recovered entity accuracy: 84.64%
Recovered entity accuracy on unique test IDs: 80.41%
Recovered triple exact-match accuracy: 56.34%
Frequency-balanced triple recovery accuracy: 74.37%

seed = 36

Output dimension: 200
Recovered entity accuracy: 79.75%
Recovered entity accuracy on unique test IDs: 84.02%
Recovered triple exact-match accuracy: 42.66%
Frequency-balanced triple recovery accuracy: 64.14%

seed = 35

Output dimension: 200
Recovered entity accuracy: 80.42%
Recovered entity accuracy on unique test IDs: 85.57%
Recovered triple exact-match accuracy: 45.47%
Frequency-balanced triple recovery accuracy: 65.08%

seed = 34

Output dimension: 200
Recovered entity accuracy: 84.84%
Recovered entity accuracy on unique test IDs: 84.02%

Recovered triple exact-match accuracy: 57.14%
Frequency-balanced triple recovery accuracy: 75.13%

seed = 33

Output dimension: 200
Recovered entity accuracy: 84.84%
Recovered entity accuracy on unique test IDs: 83.51%
Recovered triple exact-match accuracy: 56.14%
Frequency-balanced triple recovery accuracy: 75.25%

0dB,200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 71.90%
Recovered entity accuracy on unique test IDs: 60.31%
Recovered triple exact-match accuracy: 29.38%
Frequency-balanced triple recovery accuracy: 36.51%

seed = 41

Output dimension: 200
Recovered entity accuracy: 70.96%
Recovered entity accuracy on unique test IDs: 58.76%
Recovered triple exact-match accuracy: 29.38%
Frequency-balanced triple recovery accuracy: 36.90%

seed = 40

Output dimension: 200
Recovered entity accuracy: 72.43%
Recovered entity accuracy on unique test IDs: 63.92%
Recovered triple exact-match accuracy: 31.19%
Frequency-balanced triple recovery accuracy: 44.20%

seed = 39

Output dimension: 200
Recovered entity accuracy: 69.48%

Recovered entity accuracy on unique test IDs: 54.12%
Recovered triple exact-match accuracy: 24.35%
Frequency-balanced triple recovery accuracy: 30.97%

seed = 38

Output dimension: 200
Recovered entity accuracy: 67.20%
Recovered entity accuracy on unique test IDs: 59.79%
Recovered triple exact-match accuracy: 18.71%
Frequency-balanced triple recovery accuracy: 30.51%

seed = 37

Output dimension: 200
Recovered entity accuracy: 69.35%
Recovered entity accuracy on unique test IDs: 57.73%
Recovered triple exact-match accuracy: 28.77%
Frequency-balanced triple recovery accuracy: 34.87%

seed = 36

Output dimension: 200
Recovered entity accuracy: 64.32%
Recovered entity accuracy on unique test IDs: 57.73%
Recovered triple exact-match accuracy: 26.76%
Frequency-balanced triple recovery accuracy: 36.35%

seed = 35

Output dimension: 200
Recovered entity accuracy: 64.59%
Recovered entity accuracy on unique test IDs: 53.09%
Recovered triple exact-match accuracy: 15.49%
Frequency-balanced triple recovery accuracy: 28.09%

seed = 34

Output dimension: 200
Recovered entity accuracy: 71.43%
Recovered entity accuracy on unique test IDs: 60.82%

Recovered triple exact-match accuracy: 24.95%
Frequency-balanced triple recovery accuracy: 39.50%

seed = 33

Output dimension: 200
Recovered entity accuracy: 70.22%
Recovered entity accuracy on unique test IDs: 60.31%
Recovered triple exact-match accuracy: 25.35%
Frequency-balanced triple recovery accuracy: 36.54%

STE3消融

ste1：identity + ste3：sigmoid(t = 0.2)

10dB,200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 81.22%
Recovered entity accuracy on unique test IDs: 77.32%
Recovered triple exact-match accuracy: 48.69%
Frequency-balanced triple recovery accuracy: 63.23%

seed = 41

Output dimension: 200
Recovered entity accuracy: 82.56%
Recovered entity accuracy on unique test IDs: 76.80%
Recovered triple exact-match accuracy: 51.71%
Frequency-balanced triple recovery accuracy: 68.47%

seed = 40

Output dimension: 200
Recovered entity accuracy: 80.35%
Recovered entity accuracy on unique test IDs: 78.87%
Recovered triple exact-match accuracy: 45.47%
Frequency-balanced triple recovery accuracy: 53.20%

seed = 39

Output dimension: 200
Recovered entity accuracy: 84.44%
Recovered entity accuracy on unique test IDs: 80.93%
Recovered triple exact-match accuracy: 55.33%
Frequency-balanced triple recovery accuracy: 71.22%

seed = 38

Output dimension: 200
Recovered entity accuracy: 78.47%
Recovered entity accuracy on unique test IDs: 80.41%
Recovered triple exact-match accuracy: 41.25%
Frequency-balanced triple recovery accuracy: 60.44%

seed = 37

Output dimension: 200
Recovered entity accuracy: 81.49%
Recovered entity accuracy on unique test IDs: 77.84%
Recovered triple exact-match accuracy: 48.29%
Frequency-balanced triple recovery accuracy: 62.73%

seed = 36

Output dimension: 200
Recovered entity accuracy: 82.76%
Recovered entity accuracy on unique test IDs: 76.80%
Recovered triple exact-match accuracy: 49.90%
Frequency-balanced triple recovery accuracy: 65.94%

seed = 35

Output dimension: 200
Recovered entity accuracy: 82.63%
Recovered entity accuracy on unique test IDs: 80.41%
Recovered triple exact-match accuracy: 50.50%
Frequency-balanced triple recovery accuracy: 63.77%

seed = 34

Output dimension: 200
Recovered entity accuracy: 81.49%

Recovered entity accuracy on unique test IDs: 77.84%
Recovered triple exact-match accuracy: 47.48%
Frequency-balanced triple recovery accuracy: 62.89%

seed = 33

Output dimension: 200
Recovered entity accuracy: 84.31%
Recovered entity accuracy on unique test IDs: 81.96%
Recovered triple exact-match accuracy: 55.33%
Frequency-balanced triple recovery accuracy: 75.99%

0dB,200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 66.80%
Recovered entity accuracy on unique test IDs: 48.97%
Recovered triple exact-match accuracy: 22.94%
Frequency-balanced triple recovery accuracy: 28.51%

seed = 41

Output dimension: 200
Recovered entity accuracy: 66.73%
Recovered entity accuracy on unique test IDs: 54.12%
Recovered triple exact-match accuracy: 24.14%
Frequency-balanced triple recovery accuracy: 30.11%

seed = 40

Output dimension: 200
Recovered entity accuracy: 62.24%
Recovered entity accuracy on unique test IDs: 54.64%
Recovered triple exact-match accuracy: 21.73%
Frequency-balanced triple recovery accuracy: 32.75%

seed = 39

Output dimension: 200
Recovered entity accuracy: 59.36%
Recovered entity accuracy on unique test IDs: 52.58%

Recovered triple exact-match accuracy: 21.53%
Frequency-balanced triple recovery accuracy: 29.47%

seed = 38

Output dimension: 200
Recovered entity accuracy: 67.00%
Recovered entity accuracy on unique test IDs: 59.28%
Recovered triple exact-match accuracy: 26.76%
Frequency-balanced triple recovery accuracy: 37.63%

seed = 37

Output dimension: 200
Recovered entity accuracy: 66.80%
Recovered entity accuracy on unique test IDs: 56.70%
Recovered triple exact-match accuracy: 17.71%
Frequency-balanced triple recovery accuracy: 28.29%

seed = 36

Output dimension: 200
Recovered entity accuracy: 70.96%
Recovered entity accuracy on unique test IDs: 57.22%
Recovered triple exact-match accuracy: 28.57%
Frequency-balanced triple recovery accuracy: 34.25%

seed = 35

Output dimension: 200
Recovered entity accuracy: 66.33%
Recovered entity accuracy on unique test IDs: 53.09%
Recovered triple exact-match accuracy: 29.78%
Frequency-balanced triple recovery accuracy: 38.21%

seed = 34

Output dimension: 200
Recovered entity accuracy: 69.28%
Recovered entity accuracy on unique test IDs: 59.28%
Recovered triple exact-match accuracy: 25.55%
Frequency-balanced triple recovery accuracy: 35.51%

seed = 33

Output dimension: 200
Recovered entity accuracy: 61.50%
Recovered entity accuracy on unique test IDs: 58.25%
Recovered triple exact-match accuracy: 16.70%
Frequency-balanced triple recovery accuracy: 31.34%

STE1:sigmoid(t = 0.1) + STE3:sigmoid(t = 0.2)

10dB,200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 84.71%
Recovered entity accuracy on unique test IDs: 85.57%
Recovered triple exact-match accuracy: 55.33%
Frequency-balanced triple recovery accuracy: 72.59%

seed = 41

Output dimension: 200
Recovered entity accuracy: 85.71%
Recovered entity accuracy on unique test IDs: 87.11%
Recovered triple exact-match accuracy: 57.95%
Frequency-balanced triple recovery accuracy: 78.00%

seed = 40

Output dimension: 200
Recovered entity accuracy: 97.65%
Recovered entity accuracy on unique test IDs: 88.66%
Recovered triple exact-match accuracy: 93.36%
Frequency-balanced triple recovery accuracy: 92.14%

seed = 39

Output dimension: 200
Recovered entity accuracy: 97.32%
Recovered entity accuracy on unique test IDs: 89.69%
Recovered triple exact-match accuracy: 92.35%
Frequency-balanced triple recovery accuracy: 91.94%

seed = 38

Output dimension: 200
Recovered entity accuracy: 97.65%
Recovered entity accuracy on unique test IDs: 90.72%
Recovered triple exact-match accuracy: 92.96%
Frequency-balanced triple recovery accuracy: 91.12%

seed = 37

Output dimension: 200
Recovered entity accuracy: 85.45%
Recovered entity accuracy on unique test IDs: 89.69%
Recovered triple exact-match accuracy: 57.95%
Frequency-balanced triple recovery accuracy: 78.68%

seed = 36

Output dimension: 200
Recovered entity accuracy: 86.05%
Recovered entity accuracy on unique test IDs: 87.63%
Recovered triple exact-match accuracy: 59.56%
Frequency-balanced triple recovery accuracy: 82.65%

seed = 35

Output dimension: 200
Recovered entity accuracy: 86.52%
Recovered entity accuracy on unique test IDs: 86.08%
Recovered triple exact-match accuracy: 60.36%
Frequency-balanced triple recovery accuracy: 81.92%

seed = 34

Output dimension: 200
Recovered entity accuracy: 86.52%
Recovered entity accuracy on unique test IDs: 86.08%
Recovered triple exact-match accuracy: 60.36%
Frequency-balanced triple recovery accuracy: 81.92%

seed = 33

Output dimension: 200
Recovered entity accuracy: 96.58%

Recovered entity accuracy on unique test IDs: 86.60%
Recovered triple exact-match accuracy: 89.94%
Frequency-balanced triple recovery accuracy: 85.99%

0dB,200dim

seed = 42

Output dimension: 200
Recovered entity accuracy: 75.72%
Recovered entity accuracy on unique test IDs: 67.53%
Recovered triple exact-match accuracy: 35.41%
Frequency-balanced triple recovery accuracy: 53.20%

seed = 41

Output dimension: 200
Recovered entity accuracy: 75.45%
Recovered entity accuracy on unique test IDs: 67.53%
Recovered triple exact-match accuracy: 35.81%
Frequency-balanced triple recovery accuracy: 43.44%

seed = 40

Output dimension: 200
Recovered entity accuracy: 75.45%
Recovered entity accuracy on unique test IDs: 67.53%
Recovered triple exact-match accuracy: 35.81%
Frequency-balanced triple recovery accuracy: 43.44%

seed = 39

Output dimension: 200
Recovered entity accuracy: 74.25%
Recovered entity accuracy on unique test IDs: 62.89%
Recovered triple exact-match accuracy: 34.00%
Frequency-balanced triple recovery accuracy: 41.09%

seed = 38

Output dimension: 200
Recovered entity accuracy: 73.98%
Recovered entity accuracy on unique test IDs: 65.98%

Recovered triple exact-match accuracy: 35.21%
Frequency-balanced triple recovery accuracy: 43.34%

seed = 37

Output dimension: 200
Recovered entity accuracy: 75.59%
Recovered entity accuracy on unique test IDs: 65.98%
Recovered triple exact-match accuracy: 36.22%
Frequency-balanced triple recovery accuracy: 54.27%

seed = 36

Output dimension: 200
Recovered entity accuracy: 75.59%
Recovered entity accuracy on unique test IDs: 64.95%
Recovered triple exact-match accuracy: 35.81%
Frequency-balanced triple recovery accuracy: 44.37%

seed = 35

Output dimension: 200
Recovered entity accuracy: 78.27%
Recovered entity accuracy on unique test IDs: 63.92%
Recovered triple exact-match accuracy: 41.85%
Frequency-balanced triple recovery accuracy: 47.99%

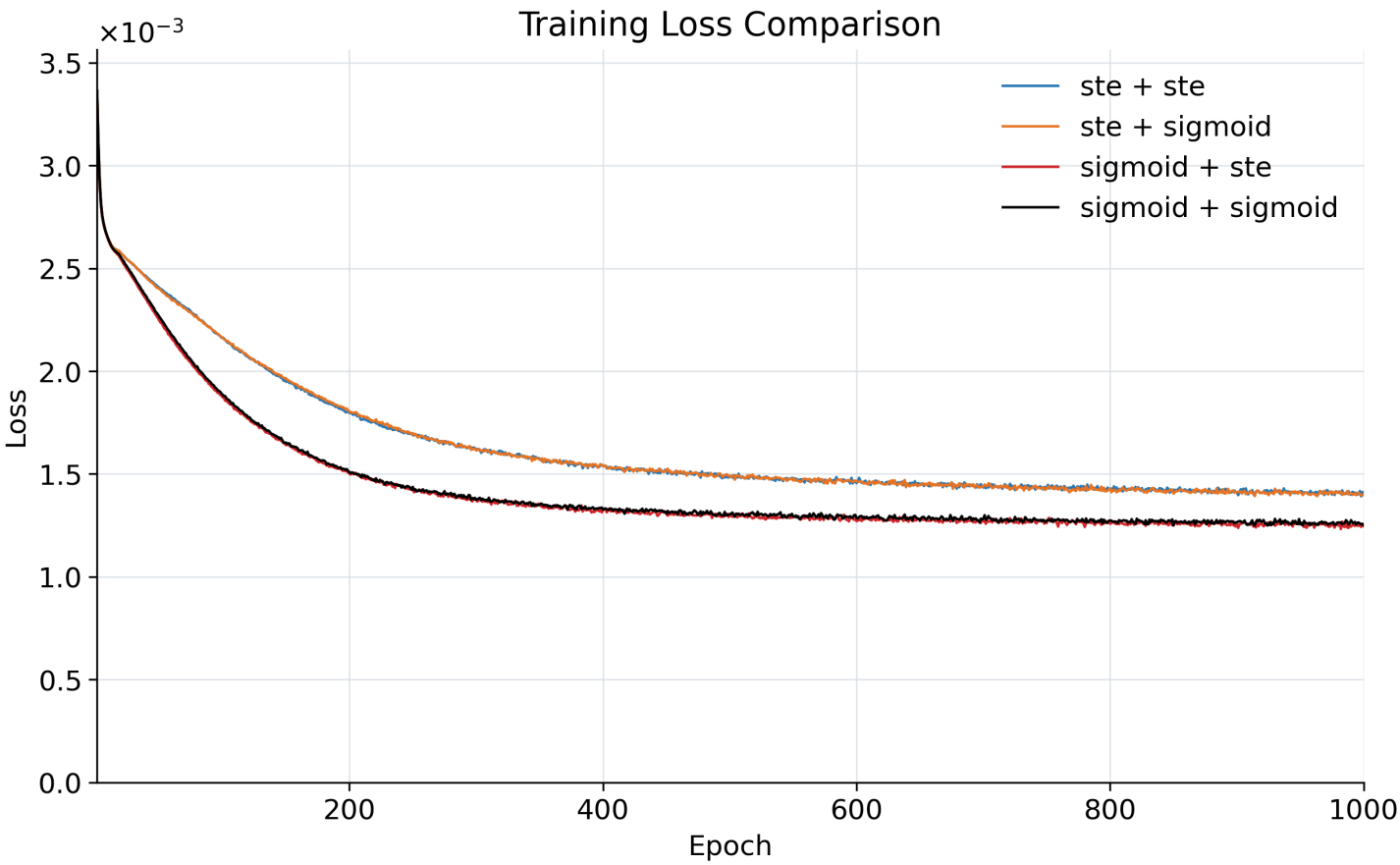
seed = 34

Output dimension: 200
Recovered entity accuracy: 69.28%
Recovered entity accuracy on unique test IDs: 64.95%
Recovered triple exact-match accuracy: 23.94%
Frequency-balanced triple recovery accuracy: 41.09%

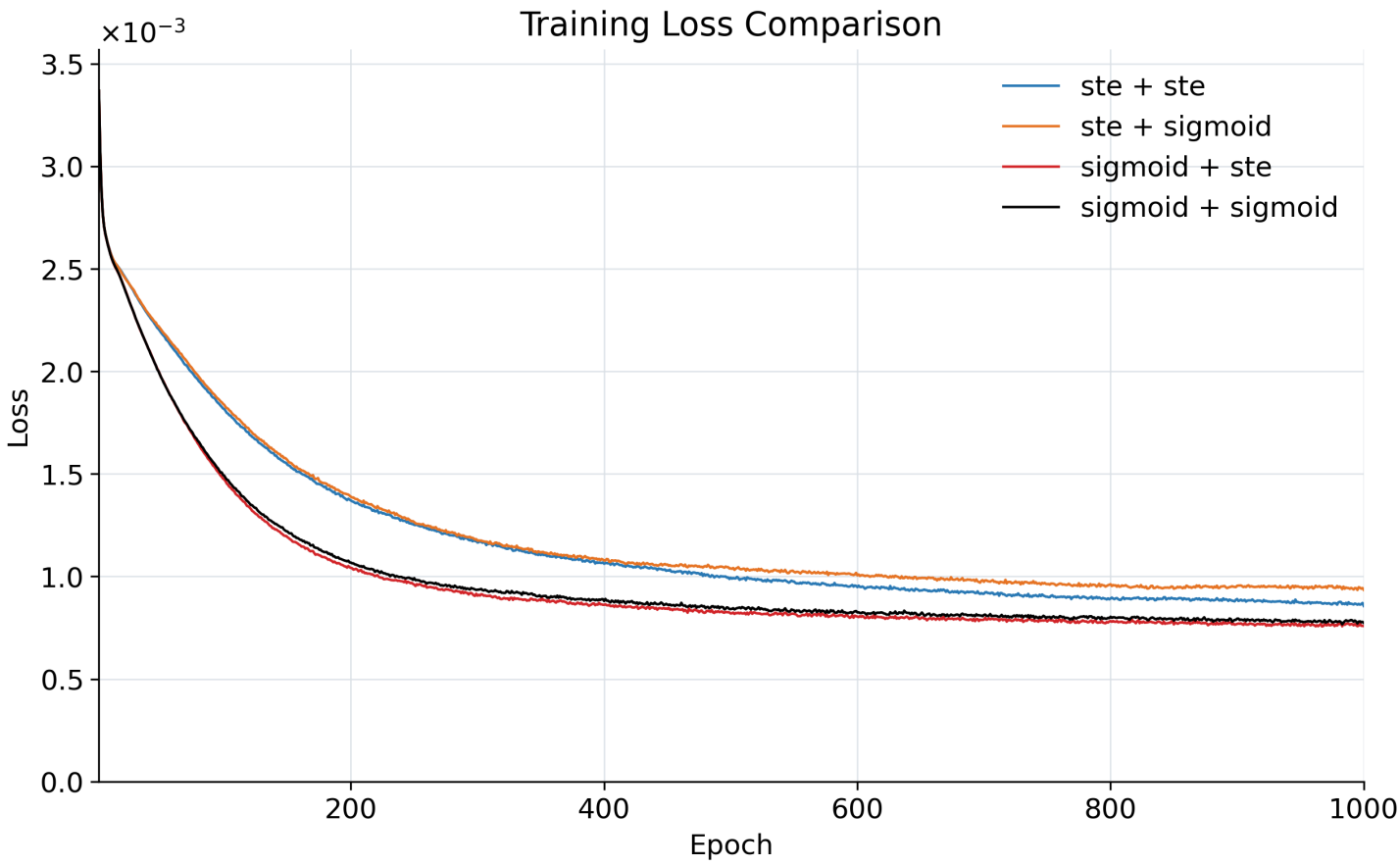
seed = 33

Output dimension: 200
Recovered entity accuracy: 76.06%
Recovered entity accuracy on unique test IDs: 68.56%
Recovered triple exact-match accuracy: 36.42%
Frequency-balanced triple recovery accuracy: 48.78%

0dB,200dim


Training Loss Comparison
×10⁻³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Loss
200
400
600
800
1000
Epoch
ste + ste
ste + sigmoid
sigmoid + ste
sigmoid + sigmoid

10dB,200dim


Training Loss Comparison
×10⁻³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Loss
200
400
600
800
1000
Epoch
ste + ste
ste + sigmoid
sigmoid + ste
sigmoid + sigmoid